

Department of Computer Science and Engineering
Faculty of Engineering & Technology, Gurukula Kangri Deemed to be University, Haridwar
Second Sessional Exam
Subject: Artificial Intelligence
Paper Code: BCE-O544

Time: 1 Hr

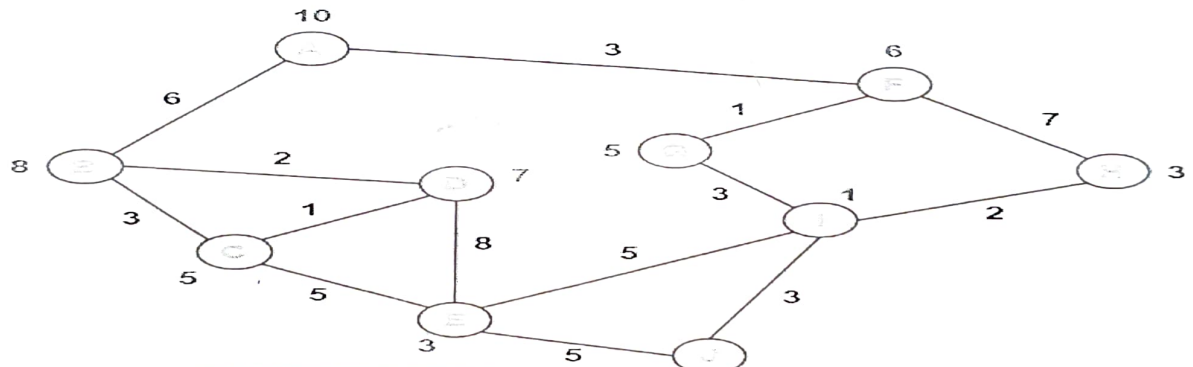
MM: 20

Note: Answer any two questions from Section A (each question carries 5 marks).
Answer any one question from Section B (each question carries 10 marks).

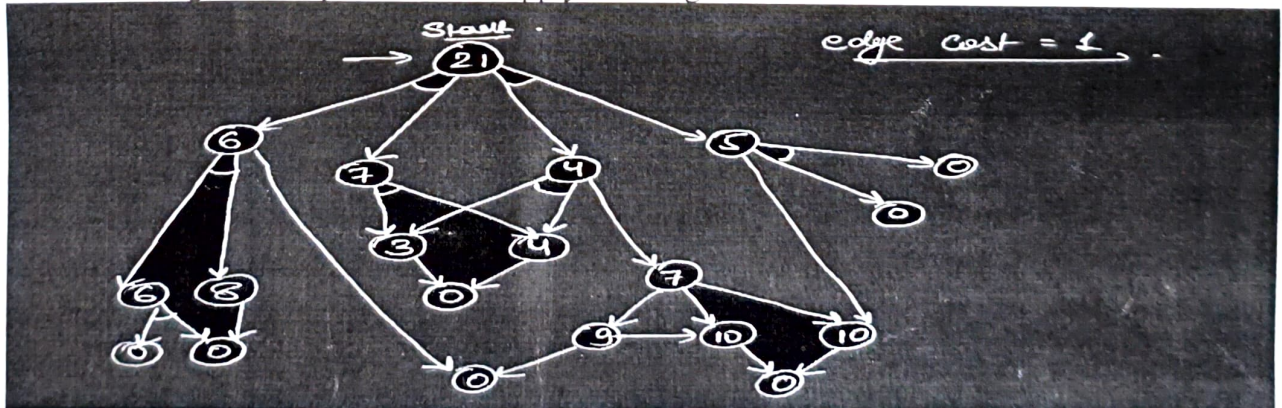
Q.No.	Section A (Questions of 5 marks each)
Q1.	a) Show that $(p \rightarrow r) \wedge (q \rightarrow r)$ and $(p \vee q) \rightarrow r$ are logically equivalent. b) prove $\neg p \rightarrow (q \rightarrow r)$ and $q \rightarrow (p \vee r)$ are equivalent?
Q2.	Check the consistency of the following Statements a) A: If it is raining then the Ground is wet. B: It is Raining or It is Snowing. C: The Ground is not wet. b) A: If it is Sunny then it is warm. B: it is Sunny or it is snowing. C: it is not warm.
Q3.	P : You have the fee. Q : You miss the final examination. r : You pass the course. a) $(p \rightarrow \neg r) \vee (q \rightarrow \neg r)$ b) $(p \wedge q) \vee (\neg q \wedge \neg r)$

Section B (Questions of 10 marks each)

- Q1. Consider the following graph-
The numbers written on edges represent the distance between the nodes.
The numbers written on nodes represent the heuristic value.
Find the most cost-effective path to reach from start state A to final state J using A* Algorithm.



- Q2. Consider the given Graph below and apply AO* algorithm on it.

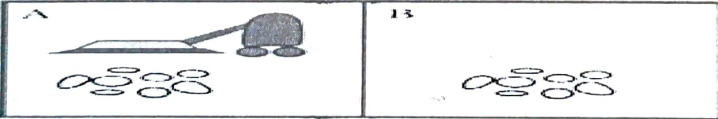
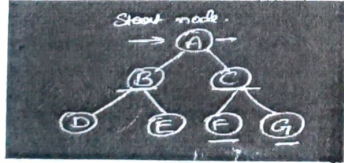


Each node in the graph is assigned a heuristic value, denoted as $h(n)$, and the edge length is considered as 1.

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Note: Answer any three questions from Section A (each question carries 4 marks). Answer any one question from Section B (each question carries 8 marks).

Q.No.	Section A (Questions of 4 marks each)	Keyword	CO	BT
Q1.	Define Artificial Intelligence and explain its importance in real-world applications.	Remembering	(CO2)	BT1
Q2.	Define Types of Agents in Artificial Intelligence.	Understanding	(CO2)	BT2
Q3.	How do Agents interact with their Environment in AI? Explain.	Understanding	(CO2)	BT2
Q4.	In the context of AI, what is meant by search? Write about the parameters of search.	Applying	(CO2)	BT3
	Section B (Questions of 8 marks each)			
Q1.	Consider the Vacuum Cleaner World Problem. Draw its state space representation and explain how an agent will work in this environment 	Applying	(CO3)	BT3
Q2.	Apply BFS (Breadth First Search) and DFS (Depth First Search) to find the path from Start node A to Goal node G in the following graph. Show step-by-step traversal. 	Evaluating	(CO4)	BT5